

Synced Drift: A sport which harmonises movement from multiple bodies to transcend distance and ability

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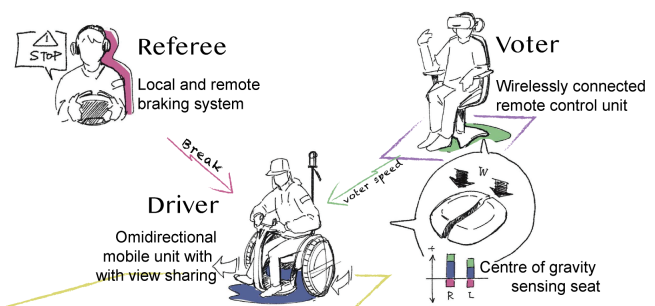


Figure 1: Overview of Synced Drift a new inclusive sport that allows two players to jointly control an omidirectional mobile unit using a systems that detects changes in the position of the centre of mass on two wirelessly connected seats

ABSTRACT

“Synced Drift” provides a new inclusive sport experience that allows two individuals to play a drifting race together while sharing a single body. Synced drift leverages a system consisting of a large-diameter omni-wheel mechanism enabling omnidirectional movement, two sensing seats that detects the user’s center of gravity shift, a mobile vehicle, and a viewpoint sharing system. The movement of the mobile vehicle is achieved by proportional control of the body movements acquired by the two sensing seats, one placed on the vehicle and one remote, from two or more individuals. The speed and direction are adjusted based on the harmonization of the inputs from the two users. This system allows people who were previously unable to participate in sports due to physical limitations, for example people with spinal cord injuries, or those with who had a stroke, to participate equally, regardless of individual impairments. Using the above system, the rules of the sport were designed and implemented. In this demonstration, the system explores the possibility of people coming together, racing and competing as one, transcending physical characteristics and physical locations.

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CCS CONCEPTS

• Human-centered computing → Accessibility technologies.

KEYWORDS

inclusive, sport, wheelchair

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1 INTRODUCTION

In 1944 Dr Ludwig Guttmann pioneered the idea of introducing sports to promote the physical and emotional recovery of war veterans [Gold and Gold 2007]. Regardless of individual characteristics, everyone had to play using a wheelchair, which helped veterans to improve their strength and endurance, but also promoted the idea of a level playing ground where everyone could compete equally, despite their role and their physical injuries [Gold and Gold 2007]. A novel way to achieve this is through the model of Superhuman sports where human augmentation technologies are used to enable equal participation and competition for all individuals [Kunze et al. 2017]. A notable example of a Superhuman sport is “SlideRift”, which combines an omnidirectional wheel system and electric assist motors [Ando et al. 2021]. However, even in this sport, there is still

no proposed solution for individuals facing upper limb impairments. Moreover, players have to be able to attend SlideRift competitions in person, which might be not feasible for those who faces barriers in leaving the house. Finally, SlideRift participants can only compete against each other, and are not able to mutually support each other to enhance their performance. To overcome these limitations, we designed a comprehensive system that integrates three components: (1) the SlideRift omnidirectional wheel-mechanism, enhancing drift capability and compensating for deficiencies in accumulated motor function; (2) a pair of interconnected sensing seats capable of detecting the user's centre of gravity, akin to the Remote Furniture [Fujimura 2004], facilitating effective mutual communication; and (3) a viewpoint-sharing system that encourages active participation from others. Harnessing the unique features of this system, we conceptualized and introduced the novel inclusive sport 'Synced Drift.' In this sport, two participants can collaboratively race, sharing a unified kinesthetic experience, irrespective of their geographical locations or individual attributes.

2 SYSTEM DESIGN

The system combines a custom-made omni-directional frame with large-diameter rear wheels featuring 56 small rubber wheels perpendicular to the circumference enabling the mobile unit to freely move along a plane parallel to the ground. Four load cells are installed on the Driver seating surface to estimate the user's center of gravity position, to direct the action of the hub motors matching the movement of the user's body for intuitive control (Figure1). A second wirelessly connected Voter remote control seat is equipped with the same mechanism, and the final movement of the wheelchair is computed by harmonising the input from the two seats. Finally, wireless communication between a camera mounted on the wheelchair and a display screen positioned in front of the remote control seat, enables real-time transmission of visual information with low latency between the vehicle and the remote operator (Figure1). For safety reasons the system is equipped with two shut down buttons that can be activated to disengage the motors, one placed on the wheelchair itself, to be controlled by the user, and a remote one which is controlled by the referee if needed. A visual representation of the centre of gravity of both Driver and Voter, is also provided. The shared position information and visual cues indicating the desired directions enable easier collaboration between participants. Although the current set up features two seats as an application, the system allows for operation based on the transmission of different bodily information beyond the center of gravity. For example, if it is difficult for an individual to leave their bed, the system can be adapted to sense the position coordinates of their head.

3 USABILITY TEST

The system has been tested with With 6 individuals who use wheelchairs for mobility as a result of different conditions including cerebral hemorrhage, spinal cord infarction, cerebral palsy, ventricular leukomalacia, spina bifida, hydrocephalus complications, and spinal cord injuries, together with their chosen partners who were non-wheelchair users. Results from the study showed that the system is safe and intuitive to use. All participants were able to enjoy the race and successfully complete the course both Driver and Voter

roles, regardless of their characteristics. During interviews users highlighted the potential of the system to promote opportunities from shared fun and support across mixed abilities and even remote locations. Based on the suggestions from participants we developed the concept of the Synced Drift sport.

4 SYNCED DRIFT RULES

To play Synced Drift two players are needed, each assigned one of the following roles:

- The Driver: who seats on the mobile unit.
- The Voter: who seats on the remote control seat placed in front of a display, showing the point of view of the camera mounted on the mobile unit.

After they have taken position according to their assigned roles, participants need to complete three consecutive laps around the designed course, which includes both clockwise and counterclockwise drifts, the time of each lap (measured as the distance between Start and End line) will be monitored, and the best lap time for each team will be entered in the leader-board.

The course features a Run-Up-Area and a Stop-Area which are not timed to make sure that the first lap time is unaffected by the need to start the wheelchair from a static position and the third lap time is not affected by the need to come to a complete stop.

At SIGGRAPH 2023, we will present a demonstration in the form of a competition available to all conference attendees. Participants, will be allotted an initial practice session of up to 3 minutes. During this period, they will learn to maneuver the mobile unit in all directions by collectively shifting their center of gravity. Upon concluding the practice round, Synced Drift competitors will engage in pairs, adhering to the prescribed official rules. Our proposition is centered around sports that emphasize collaborative harmonization over individual physical prowess. Our aim is to pioneer a sporting paradigm that fosters enjoyment for all participants.

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